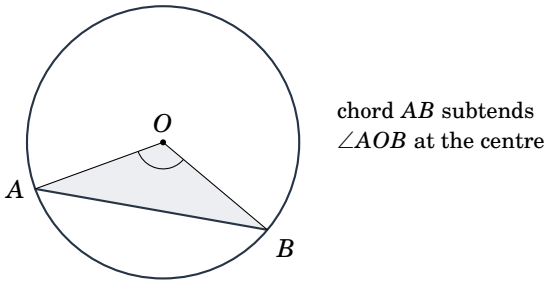
Circles

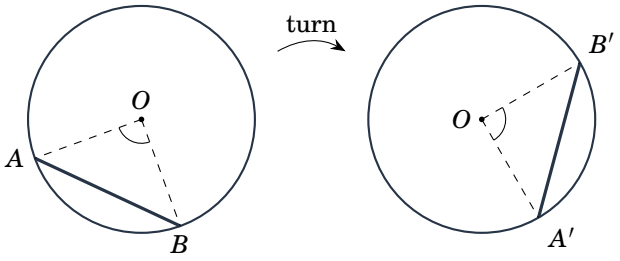
Class 1

1 Chords and the centre

A **chord** joins two points of a circle. A chord through the centre is a **diameter**. Join the ends A and B of a chord to the centre O. The chord AB and the radii OA and OB form $\angle AOB$. We say that the chord AB **subtends** $\angle AOB$ at the centre.



Tie a thread tightly between two points on the rim of a bicycle wheel. The thread is a chord of the circle. Now turn the wheel. The thread moves to a new position, but its length does not change.



Does the angle that the thread subtends at the centre change? Solve the problems below before you decide.

2 Equal chords and equal angles at the centre

Solve first.

P1. A circle has centre O and radius 5 cm. Chords AB and CD are each 6 cm long. Draw a rough diagram and join OA, OB, OC and OD.

(a) Write the three pairs of equal sides in $\triangle AOB$ and $\triangle COD$.

(b) Are the two triangles congruent? Name the congruence rule.

(c) What can you say about $\angle AOB$ and $\angle COD$?

P2. In another circle with centre O and radius 5 cm, $\angle POQ = \angle ROS = 60°$.

(a) What kind of triangle is $\triangle POQ$? Give a reason.

(b) Find the lengths of the chords PQ and RS.

(c) What do you notice about the two chords?

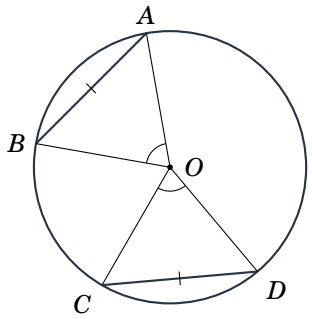
P3. In P2 the angle 60° made the calculation easy. Now let $\angle POQ = \angle ROS = 50°$ instead.

(a) You cannot calculate PQ yet. Write two pairs of equal sides and one pair of equal angles in $\triangle POQ$ and $\triangle ROS$.

(b) Are the two triangles congruent? Name the rule.

(c) What follows for the chords PQ and RS?

Theorem 1 (NCERT Theorem 2). *Equal chords of a circle subtend equal angles at the centre.*



Why it works.

Given: chords AB and CD of a circle with centre O, and $AB = CD$.
To show: $\angle AOB = \angle COD$.

1. In $\triangle AOB$ and $\triangle COD$: $OA = OC$ and $OB = OD$ (radii), and $AB = CD$ (given).
2. So $\triangle AOB \cong \triangle COD$ (SSS).
3. Hence $\angle AOB = \angle COD$ (corresponding parts of congruent triangles).

Theorem 2 (NCERT Theorem 3). *Chords of a circle that subtend equal angles at the centre are equal.*

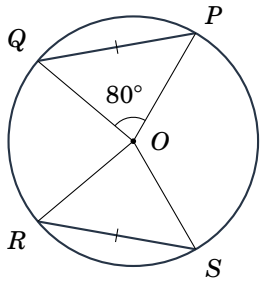
Why it works.

Given: chords AB and CD of a circle with centre O, and $\angle AOB = \angle COD$ (same figure).
To show: $AB = CD$.

1. In $\triangle AOB$ and $\triangle COD$: $OA = OC$ (radii), $\angle AOB = \angle COD$ (given) and $OB = OD$ (radii).
2. So $\triangle AOB \cong \triangle COD$ (SAS).
3. Hence $AB = CD$.

Theorem 2 is the **converse** of Theorem 1: the given fact and the conclusion change places.

Example 1. In a circle with centre O, chords PQ and RS are equal and $\angle POQ = 80°$. Find $\angle ROS$ and $\angle ORS$.

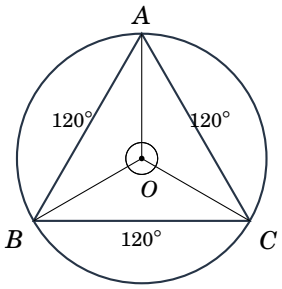


Given: $PQ = RS$ and $\angle POQ = 80°$.
To find: $\angle ROS$ and $\angle ORS$.

1. $\angle ROS = \angle POQ = 80°$ (equal chords subtend equal angles at the centre, Theorem 1).
2. $OR = OS$ (radii), so $\triangle ORS$ is isosceles and $\angle ORS = \angle OSR$.
3. $\angle ORS + \angle OSR = 180° - 80° = 100°$ (angle sum of a triangle), so $\angle ORS = 50°$.

Answer: $\angle ROS = 80°$ and $\angle ORS = 50°$.

Example 2. A, B and C are points on a circle with centre O such that $\angle AOB = \angle BOC = 120°$. Show that $\triangle ABC$ is equilateral.



Given: $\angle AOB = \angle BOC = 120°$.
To show: $AB = BC = CA$.

1. The three angles at O make a complete angle, so $\angle COA = 360° - 120° - 120° = 120°$.
2. The chords AB, BC and CA subtend equal angles at the centre, so $AB = BC = CA$ (Theorem 2).

Answer: $\triangle ABC$ is equilateral.

Exercise 1A

1. In a circle with centre O, chords AB and CD are equal and $\angle AOB = 70°$. Find $\angle COD$ and $\angle OCD$. Give a reason for each step.
2. In a circle with centre O, chord $PQ = 7$ cm and $\angle POQ = 90°$. Another chord RS also subtends a right angle at O. Find RS and $\angle ORS$.
3. Show that the triangle formed by a chord and the centre of the circle is isosceles. Then show that two such triangles in the same circle with equal bases are congruent.
4. Draw a circle of radius 3 cm and a circle of radius 5 cm. In each, draw a chord of length 4 cm and join its ends to the centre. Measure the two angles at the centres. Rani says, "Equal chords subtend equal angles at the centre, so these angles must be equal." Is she right? Explain.

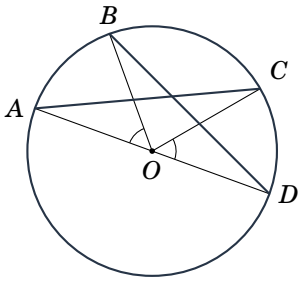
5. A regular hexagon $ABCDEF$ is drawn with all its vertices on a circle with centre O and radius 4 cm.

   (a) Why do the six sides subtend equal angles at O? Find each angle.

   (b) What kind of triangle is $\triangle OAB$? Find the perimeter of the hexagon.

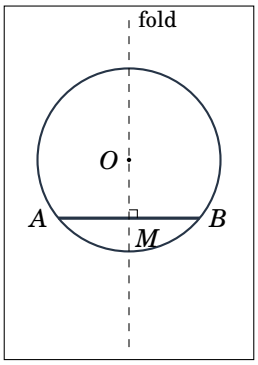
6. The tip of the minute hand of a wall clock is at 12 at 10:00 and at 4 at 10:20. Later it is at 3 at 10:15 and at 7 at 10:35. Explain why the straight-line distance between the first two positions of the tip equals the distance between the last two.

7. A, B, C and D lie in this order on a circle with centre O, and $\angle AOB = \angle COD$. Show that $AC = BD$.



3 The line from the centre to the midpoint of a chord

Draw a circle of radius about 6 cm on an A4 sheet with your compass, and mark its centre O. Draw any chord AB that is not a diameter. Hold the sheet against a window and fold it so that A falls exactly on B. Open the sheet.



Does the crease pass through O? What angle does it make with AB? Where does it cut AB? Now explain what you saw.

Solve first.

P4. A circle has centre O and radius 5 cm. Chord AB is 8 cm long and M is the midpoint of AB. Join OA, OB and OM.

   (a) Show that $\triangle OMA \cong \triangle OMB$. Name the rule.

   (b) $\angle OMA + \angle OMB = ?$ Use this to find $\angle OMA$.

   (c) Find the length OM.

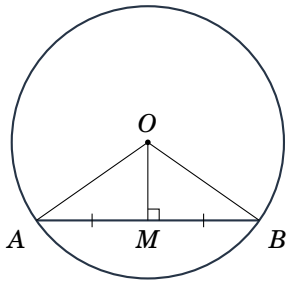
P5. A circle has centre O and radius 10 cm. PQ is a chord, and $ON \perp PQ$ with $ON = 6$ cm. Join OP and OQ.

(a) Name the two right triangles. Are they congruent? Name the rule.

(b) Find PN and NQ.

(c) What do you notice about the point N?

Theorem 3 (NCERT Theorem 4). *The line joining the centre of a circle and the midpoint of a chord of the circle is perpendicular to the chord.*



Why it works.

Given: chord AB of a circle with centre O, and M the midpoint of AB.
To show: $OM \perp AB$.

1. In $\triangle OMA$ and $\triangle OMB$: $AM = BM$ (given), $OA = OB$ (radii) and OM is common.
2. So $\triangle OMA \cong \triangle OMB$ (SSS), and $\angle OMA = \angle OMB$.
3. These angles form a linear pair, so their sum is 180° and each is 90°.

Theorem 4 (NCERT Theorem 5). *The perpendicular from the centre of a circle to a chord of the circle bisects the chord.*

Why it works.

Given: chord AB of a circle with centre O, and $OM \perp AB$ with M on AB (same figure).
To show: $AM = BM$.

1. In $\triangle OMA$ and $\triangle OMB$: $\angle OMA = \angle OMB = 90^\circ$, $OA = OB$ (radii) and OM is common.
2. So $\triangle OMA \cong \triangle OMB$ (RHS).
3. Hence $AM = BM$.

Your fold showed one more fact. The crease is the **perpendicular bisector** of AB: it passes through the midpoint M and is perpendicular to AB. By Theorem 3, the line OM also passes through M and is perpendicular to AB. Only one line through M is perpendicular to AB, so the crease and the line OM are the same line.

Things to Remember

The perpendicular bisector of every chord of a circle passes through the centre.

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4 The half-chord right triangle

In P4 the radius, half of the chord and the distance OM formed a right triangle. The **distance of a chord from the centre** is the length of the perpendicular from the centre to the chord.

Solve first.

P6. A chord of a circle has length c and is at distance d from the centre. The radius is r.

(a) Draw the figure and mark the right triangle. Which side of this triangle is the radius? Which side has length $\dfrac{c}{2}$, and why?

(b) Write one equation connecting r, c and d.

(c) Check your equation with the numbers in P4.

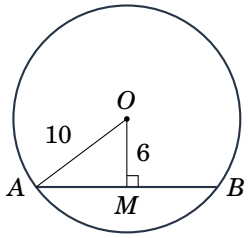
Things to Remember

For a chord of length c at distance d from the centre of a circle of radius r,

$$\left(\frac{c}{2}\right)^2 + d^2 = r^2.$$

Use *half* of the chord, never the whole chord.

Example 3. A chord of a circle of radius 10 cm is 6 cm from the centre. Find the length of the chord.

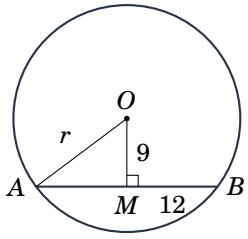


Given: radius $OA = 10$ cm, $OM \perp AB$, $OM = 6$ cm.
To find: the length of chord AB.

1. $AM = MB$ (the perpendicular from the centre bisects the chord, Theorem 4).
2. In right $\triangle OMA$, $AM^2 = OA^2 - OM^2 = 100 - 36 = 64$, so $AM = 8$ cm.
3. $AB = 2 \times AM = 16$ cm.

Answer: the chord is 16 cm long.

Example 4. A chord of length 24 cm is 9 cm from the centre of a circle. Find the radius of the circle.

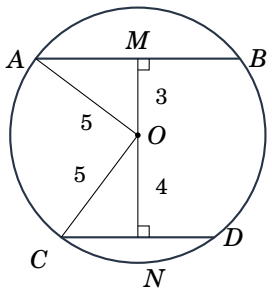


Given: $AB = 24$ cm, $OM \perp AB$, $OM = 9$ cm.
To find: the radius OA.

1. $AM = \dfrac{24}{2} = 12$ cm (Theorem 4).
2. In right $\triangle OMA$, $OA^2 = AM^2 + OM^2 = 144 + 81 = 225$, so $OA = 15$ cm.

Answer: the radius is 15 cm.

Example 5. Two parallel chords of lengths 6 cm and 8 cm lie on opposite sides of the centre of a circle of radius 5 cm. Find the distance between the midpoints of the chords.



Given: $AB \parallel CD$, $AB = 8$ cm, $CD = 6$ cm, radius 5 cm, O between the chords; M, N the midpoints of AB and CD.
To find: MN.

1. $OM \perp AB$ (Theorem 3). A line perpendicular to AB is also perpendicular to the parallel chord CD, and the line through O perpendicular to CD meets CD at its midpoint N (Theorem 4). So M, O and N lie on one straight line.
2. In right $\triangle OMA$, $OM^2 = 5^2 - 4^2 = 9$, so $OM = 3$ cm.
3. In right $\triangle ONC$, $ON^2 = 5^2 - 3^2 = 16$, so $ON = 4$ cm.
4. $MN = OM + ON = 3 + 4 = 7$ cm.

Answer: 7 cm.

Draw a rough figure for every question in Exercise 1B and mark all the given measurements on it.

Exercise 1B

1. The radius of a circle is 13 cm and a chord is 5 cm from the centre. Find the length of the chord.
2. A chord of length 30 cm is drawn in a circle of radius 17 cm. Find the distance of the chord from the centre.
3. The diameter of a circle is 26 cm. A chord of length 24 cm is drawn in it. Find the distance of the chord from the centre.
4. Chord AB of a circle with centre O is 48 cm long. $OM \perp AB$ and $OM = 7$ cm. Ravi writes: radius $= \sqrt{48^2 + 7^2}$. Find his mistake, and find the correct radius.
5. Draw a circle of radius 6.5 cm and a chord of length 12 cm in it. Use a set square to draw the perpendicular from the centre to the chord, and measure it. Then calculate its length and compare.

6. A round dining-table top has radius 50 cm. A carpenter fixes a straight wooden strip on the underside along a chord, 30 cm from the centre. Find the length of the strip.

7. A drainage pipe has an inner radius of 13 cm. Water flows in it, and the water surface is 24 cm wide. The water fills less than half of the pipe. Find the depth of the water.

8. True or false? Give a reason for each.

   (a) Every line through the midpoint of a chord passes through the centre.

   (b) A line through the centre that bisects a chord is always perpendicular to the chord.

9. An isosceles triangle ABC, with $AB = AC$, has all its vertices on a circle with centre O. Show that the altitude from A to BC passes through O.

10. Two parallel chords of lengths 6 cm and 8 cm lie on the same side of the centre of a circle of radius 5 cm. Find the distance between them.

11. Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre of a circle, and the distance between them is 7 cm. Find the radius of the circle.